

Session 5: Streamflow Data Assimilation in T-Route

Enable USGS streamflow data assimilation (DA) in T-Route for both the Muskingum-Cunge (MC) and Diffusive Wave (DW) schemes. The open-loop `troute.yaml` from Session 4 is reused as the starting point; the only changes required are (1) preparing the USGS timeslice observations and (2) editing the `data_assimilation_parameters` block to enable nudging.

Prepare USGS Timeslice Observations

T-Route reads observed streamflow from NWM-style timeslice NetCDF files. Extract the 2020 archive into a flat directory (no subdirectories), which is the layout T-Route expects.

Step 1: Create the timeslices directory and extract the archive

Run the following from the directory containing `usgstimeslice2020.zip`:

```
mkdir -p /workspace/data/timeslices
python -c "
import zipfile, os
target = '/workspace/data/timeslices/'
os.makedirs(target, exist_ok=True)
with zipfile.ZipFile('usgstimeslice2020.zip', 'r') as z:
    for f in z.namelist():
        filename = os.path.basename(f)
        if filename:
            source = z.open(f)
            target_path = os.path.join(target, filename)
            with open(target_path, 'wb') as out:
                out.write(source.read())
print('Done')
"
```

The `os.path.basename(f)` step strips any internal folder structure from the zip so all timeslice files land directly in `/workspace/data/timeslices/`.

Step 2: Verify extraction

List the first few files to confirm the extraction worked:

```
ls /workspace/data/timeslices/ | head -5
```

Muskingum-Cunge (MC) with Data Assimilation

Starting from the open-loop MC `troute.yaml` (Session 4), replace the `data_assimilation_parameters` block with the DA-enabled version below. It is also recommended to update `stream_output_directory` to a DA-specific path (e.g., `/workspace/data/DA/MC/Output/`) so DA results are kept separate from open-loop results.

Step 1: Enable streamflow DA in the MC configuration

```
data_assimilation_parameters:
  #   #-----
  usgs_timeslices_folder    : /workspace/data/timeslices/
  #   usace_timeslices_folder : #usace_TimeSlice/
```

```

timeslice_lookback_hours : 0
#   qc_threshold           : #1
streamflow_da:
#-----
streamflow_nudging: True
diffusive_streamflow_nudging: False

```

Parameter reference:

- `usgs_timeslices_folder` : path to the flat directory of USGS timeslice files prepared above.
- `timeslice_lookback_hours`: 0 : do not look back beyond the current timestep when searching for observations.
- `streamflow_nudging`: True : enable nudging on Muskingum-Cunge reaches (all reaches under MC-only runs).
- `diffusive_streamflow_nudging`: False : DW nudging is off for MC-only runs.

Step 2: Run T-Route with DA (MC)

Launch the routing engine with the same command used for the open-loop run:

```
python -m nwm_routing -f -V4 troute.yaml
```

Diffusive Wave (DW) with Data Assimilation

For the DW run with DA, keep the `hybrid_parameters` block from Session 4 (unchanged), keep `forcing_parameters` as-is, and set both nudging flags to True in `data_assimilation_parameters`. As before, direct output to a DA-specific path (e.g., `/workspace/data/DA/DW/Output/`).

Step 1: Configure the DW YAML with DA enabled

Relevant sections of the DW `troute.yaml` for the DA run:

```

hybrid_parameters:
  run_hybrid_routing: True
  diffusive_domain : /workspace/data/Openloop/DW/Domain/Domain.yaml
  use_natl_xsections: True
  topobathy_domain : /workspace/data/Openloop/DW/Domain/Mainstem.parquet
#   run_refactored_network: False
#   refactored_domain: # domain/unit_test_noRS/refactored_coastal_domain.yaml
#   refactored_topobathy_domain: # domain/refac_final_diffusive_natural_xs.nc
#   coastal_boundary_domain: domain/coastal_boundary_domain.yaml
forcing_parameters:
#-----
  qts_subdivisions: 12
  dt: 300 # [sec] 300 == 5 minutes
  qlat_input_folder: /workspace/data/Coralville2/outputs/ngen/
  qlat_file_pattern_filter: "cat-*"
  nts: 105408.0 #288 for 1day
  max_loop_size: 105408.0 # [number of timesteps]
data_assimilation_parameters:
#   #-----
  usgs_timeslices_folder : /workspace/data/timeslices
#   usace_timeslices_folder : #usace_TimeSlice/

```

```
timeslice_lookback_hours : 0
#   qc_threshold          : #1
streamflow_da:
#-----
streamflow_nudging: True
diffusive_streamflow_nudging: True
```

Key delta versus the MC DA configuration: `diffusive_streamflow_nudging: True`. This applies observation nudging to the mainstem reaches solved by the diffusive kernel, in addition to the MC nudging applied to the remaining tributary reaches.

Step 2: Run T-Route with DA (DW)

Launch the routing engine with the same command as before:

```
python -m nwm_routing -f -V4 troute.yaml
```